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Project Title:	Gravity and Granular Materials		
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Program/Discipline:	FLUID PHYSICS		
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Comments:	NOTE: PI Robert Behringer is deceased, as of July 2018.		
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No. of Master's Candidates:	4	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Flight Assignment:	Note- grant renewed for 4 years/new NAG#		
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COI Name (Institution):	Kondic, Lou (New Jersey Institute of Technology)		
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	The initial task consisted of a detailed characterization of the dynamics of shaken granular materials. Experiments identified important behavior and flow properties for shaken granular materials. Shaking was carried out both vertically and horizontally. The vertical component of shaking provided a means to modify gravity, and the horizontal component provided oscillating shear stress. This technique provides an excellent method to explore concepts of granular friction and failure. A detailed account of this work has appeared in Physical Review E, and in Europhysics Letters. More recently the task has focused on the properties of slowly sheared dense granular materials. Using a novel apparatus that allows characterization of all aspects of this process in 2 dimensions, we have discovered a novel transition that likely plays a key role in the failure of granular materials under applied stress. An initial account of this work has appeared in Physical Review Letters, and a more detailed account has appeared in Springer Lecture Notes in Physics and in a manuscript that is in preparation. This is a very complex system, and additional work is now being focused on		

Task Description:	understanding its properties. These explorations have recently extended to a much larger 2D system. The task was extend to include a flight mission using a 3D system. In this regard, a key point is that on earth gravity masks the true dynamical properties of both rapidly and slowly sheared granular systems. At this stage, however, the flight aspect of this project is no longer active, due to changes in NASA goals and missions. An additional key task, intimately related to the flight experiment, is a set of Molecular Dynamics (MD) studies in both 2D and 3D. These studies are of considerable scientific importance in their own right, and provide key insights into what might be expected in a microgravity environment. The task has been broadened over the past year by the addition of two new related projects. one of these is a 3D granular Couette experiment that is effectively a prototype for a flight experiment. The other is an experiment to probe the interaction of granular materials with a frictional boundary. This project has also focused on characterizing fluctuations and diffusion in 2D sheared systems, both experimentally and by MD simulation. Details of these results have recently appeared in Physical Review Letters, Physical Review E, European Physical Journal E, and Europhysics Letters. One final aspect has been the characterization of meteor-like impacts through the novel use of photoelastic materails. This project entails the study of the properties of granular materials. The presence of gravity plays a strong role in how granular materials behave. In some instances, it also obscures fundamental behavior that could be much better understood in the presence of reduced gravity. The objectives of this work are to 1) understand the role that gravity plays in certain granular systems, such as sheared and shaken systems, and 2) to uncover basic science that governs the way granular materials behave and that is obscured ordinarily by the presence of gravity, 3) from the increased understanding from 1 and 2, increase our ability to predict and control granular flows, thus leading to significant savings in technical and commercial processing industries in the US and abroad, and 4) provide insight into granular materials handling in reduced-gravity situations. 5) In recognition of the initiative to explore the Moon and Mars, we are particularly focusing on a) improving our understanding of predictiability on granular materials, b) understanding the scope of granular processing for Lunar and Maritan exploration, and c) investigating novel systems of 2-phase granualar-gas flow that may be highly important for granular handling in the above environments. In the present year, ending in November 2003, the objectives remain the same. The current key focus of this project is on the properties of sheared granular materials. Here, the key significance of the outcome of this task will be improvements in our understanding of the fundamental dynamical properties of granular materials. It is important to appreciate that our ability to describe these materials is primitive relative to what we can do for other materials such as Newtonian liquids. The negative consequences of this state of ignorance are substantial, and involve large financial and human losses. With a new focus on granular materials in Lunar and Martian settings, we expect to provide key insights that will be essential for the success of such missions. The shaking and shearing of granular materials are particularly important as basic means to cause flow in granular materials, and hence as tools to transport and process a variety of materials, including coal, ores, food grains, pharmaceutical powders and even dog food. These processes also plays key roles in natural phenomena such as avalanches and earthquakes. Gravity plays an intrinsically important role in these cases. Looking into the future, attempts to mine on the lunar surface will almost surely involve shaking processes for granular materials. Because our understanding of granular flows is rather poor, many of the plants and devices which handle granular materials operate significantly below capacity, and fail all too frequently. The result is corresponding loss to the US economy. The expectation is that increased scientific insight obtained from this project will lead to improved predictability and efficiency in a range of technological processes.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	A deeper understanding of the properties of granular materials is essential for a broad spectrumof earth-bound tasks. Failures of devices on the Earth are all too common, and much insight is needed. The present tasks should help significantly towards this end.
	Below, I briefly discuss key activities and findings from the past research period. I then append publications, papers submitted, preprints, and talks generated from this research. This work was carried out by the PI and several other junior researchers including: Post-docs Karen Daniels and Brian Utter; Ph.D. students Trush Majmudar, Robert Hartley, Peidong Yu, and Junfei Geng; and undergraduate students Joyce Coppock and David Marks. Coutte shear experiment: a novel phase transition We have brought on-line a highly novel Couette shear apparatus that allows us to simultaneously shear a granular sample (from above) while providing energy from below by vibration. We have used this apparatus to probe the jamming/unjamming transition in dense granular materials (publication[9]). We find a fascinating transition from a crystallized state to a disordered state in this sytem, where order is induced by vibration and disorder is induced by shear. There is a phase boundary between ordered and disordered states that corresponds to equal energy input per particle from shear and vibration. This transition is particularly striking because vibration, which is ordinarily associated with thermal disordering, is actually drives the ordering process. Here, it is important to note that gravity is present and likely playing an important role in these earthbound experiments. Work is continuing on this project to further understand the competing roles of energy imput, and other properties of this sytem. This experiment also serves as a model for the development of a flight experiment. In addtion, we are working closely with Prof. Lou Kondic of NJIT to understand what we might expect from a flight experiment, and to address fundamental physics issues that arise in this type of system. Basic statistical properties of granular systems Here, we have carried out several experimental projects, including tests for alternative variables or a granular temperature. First, in the context of Couette shear in a very large 2D system consisting of photoelastic particles, we measured the {\bf self-diffusion} of particles. (publications.[1,8]) In our experiments, particles lying in a plane, are sheared by an inner rotating wheel, and confined by an outer ring. As is typical of Couette shear, there is a shear band surrounding the shear wheel, where the density is reduced; also the azimuthal velocity falls off with distance, r, from

Task Progress:

the wheel as $v \propto \exp (-r/\lambda - {\cal O}(r^2))$, where $\lambda \simeq 5d$ is a measure of the width of the shear band, d being the grain diameter. Grains are advected within the shear band, but also show fluctuating, typically diffusive motion. The presence of a nonlinear velocity profile and the impenetrable boundary at the wheel lead to a complex diffusion problem. Nevertheless, we are able to match the measured diffusion coefficients well with a random walk model that reflects the mean flow and the rigid boundary. It appears that diffusion is Brownian, although actual trajectories are strongly influenced by gradients in the mean flow. This is a rather interesting result, especially in light of simulations of Radjai and Roux predicting superdiffusion.We also show that the force chain network forms so as to oppose shear, and tends to rotate the principal axes of diffusion away from the radial and azimuthal directions. We have also explored the effect of changes in the force network on transient behavior induced by reversing the direction of shear. (publication [5])

Recently, there have been a number of attempts to define an {\bf extended granular temperature} for dense materials by borrowing from the statistical mechanics of equilibrium thermal materials. This work includes the early proposal by Edwards et al.~\cite{edwards,edwards89} involving the counting of jammed configurations, studies of diffusivity and mobility in shear flow by Makse and Kurchan, work by Nagel, Liu, O'Hern and collaborators using fluctuation dissipation relations, and the use of equipartition by Kondic and Behringer~\cite{kondic_04}. For example, Makse and Kurchan~\cite{makse_02} computed an extended temperature as the ratio of diffusivity and mobility for dense shear flow.

We have carried out {\bf experimental tests of mobility and diffusion}. (publications[7,10]) Motivation for such studies included the predictions by Makse and Kurchan, experimental studies by Albert et al., and the predictions by Radjai and Roux~\cite{radjai02} already noted. The first of our experiments yielded diffusivity, D, and mobility, B, for a stirred granular system. We pushed a tracer particle slowly and at constant velocity, v, through a stirred medium to obtain the mobility, and we separately measured the diffusivity for this system. Although we observed simple Brownian diffusion in this system, we did not obtain sensible results for the mobility. Rather, the force needed to push the tracer particle through the other particles was nearly independent of v, showing only a logarithmic increase over a broad range. This behavior resembles Coulomb friction more than a molecular fluid, and precludes defining a temperature in terms of the ratio of D/B for this arrangement. We believe that the reason for the discrepancy between the Makse simulations and our experiments lies in the fact that for constant (moderate) speed pushing, the tracer % than the background fluctuation velocity, the tracer breaks force chains rather than simply taking advantage of random `thermal' fluctuations that provide opportunities for tracer motion. A pushing force much smaller than those characteristic of the force chain network may yet yield a molecular-like mobility. It would be very interesting to carry out a related experiment in which the tracer was subject to a very small constant {\em force} rather than to a constant {\em velocity}. We propose to carry out such experiments, where we will also explore the effects of changing other parameters, including the tracer size, the rate of stirring, etc.

In a rather different set of experiments, we are beginning to obtain results testing the idea of Edwards entropy or compactivity as relevant `thermodynamic' variables using a novel apparatus that consists of two different size spheres, lying on a plane and subject to shaking. There is a movable barrier between the two particle types, and as the system is shaken, the position of the barrier fluctuates/evolves in time. For such a system, Lemaitre and Bouchaud have used entropy maximum arguments to predict the position of the barrier. Results from this work are just beginning to emerge, and suggest that these predictions are valid. However, much remains to be done. This project was launched by undergrad David Marks, and will be continued by post-doc Karen Daniels. (David has recently left Duke to begin Ph.D.~studies in Physics at Stanford.)

The mobility experiment discussed above are interesting in their own right, and we studied this process in detail. (publication [10]) These experiments bear some resemblance to recent 3D studies by Albert et al.~\cite{albert_prl99,albert_prl00} in which a stiff rod was driven through glass spheres. Albert et~al.~found that the average drive force was independent of speed in contrast to our observation that the drag force grows logarithmically with the speed. Our measured logarithmic rate dependence corresponds to similar observations, discussed below, for Couette shear. The difference between our experiments and those of Albert et~al.~remains unexplained at the moment. % although both yielded a friction-like response. We were successfully able to adapt a failure model of Kahng et~al.~\cite{kahng01} to describe our drag experiments rather well. The interpretation of this model is that the drag occurs because the tracer particles do work against both existing and induced force chains that fail under sufficient force.

Regarding, we carried out extensive 2D Couette shear experiments, where we determined the force on particles using photoelasticity. (publication [15]) For at least five decades in shear rate, Ω, the mean force per particle ($\sim$pressure) grows logarithmically with Ω. This rate dependence is presumably related to that seen in sophisticated measurements of friction, and in rate-and-state models.

We extensively studied force transmission of a static system in response to a point load in 2D using photoelastic particles. (publications[2,11,14]) These studies showed two-peaked wave-like transmission for geometrically ordered packings, with less broadening of the peaks as friction was decreased. For disordered systems of moderate size, the response was elastic-like. For a system subject to a modest simple shear there was a correlated force network along the compressional direction associated with this shear and the applied point force was ``redirected' toward this strong direction. This work tested a number of models, clearly showing that parabolic models did not correctly predict the observed force response and providing motivation for a number of new theoretical developments. This work also benefited from a continuing collaboration with the Paris granular group lead by Eric Clement. Much remains to be done to help settle issues for force transmission, and we discuss additional tests below.

We carried out extensive studies of how granular materials interact frictionally with boundaries, again using 2D photoelastic techniques. In one of these experiments, particles were confined to a 2D version of a silo. (publication [12]) We determined the response of this system to an applied load from above, to a driving force from below, etc. Unlike past studies of model silos, we were able to determine forces internally, and investigate the uniformity of stresses across a horizontal slice. In this setting, a frequently invoked and very simple model due to Janssen predicts exponential saturation of pressure with depth (from the top). Although our results are qualitatively consistent with such a model, there are substantial quantitative departures. Several articles detailing these results are in draft form. A second experiment dealing with granular friction at a boundary consists of a `slider' which we pull across a vertically oriented 2D array of photoelastic disks. We simultaneously measure the force pulling the slider and the evolution of the force structures within the material. We anticipate a paper describing the results shortly.

Improvements in photoelastic techniques

An important aspect of our photoelastic studies, carried out by Ph.D. student Trushant Majmudar, has been to develop algorithms and computer code to extract vector forces at contacts between grains (publication [14]). In previous work, we used an empirical calibration relating the gradients of intensity in a photoelastic image to the force on a particle. The methods that we now have implemented can automatically process an image, compute the stress field within each particle, and determine the vector forces at contacts. We can now process images quickly and with high reliability.

Meteor impacts

A recent novel topic is impacts of objects such as meteors on soils. This issue is also quite relevant to Lunar exploration. Real meteors are massive and release large amounts of energy on impact, but some essential aspects of the physics can be probed in controlled experiments. A number of novel studies have considered the dynamics of lab-scale impact events. We have explored the effect of impacts using 2D particles and high-speed video as a novel technique to directly watch energy flow during the impact of a large intruding particle (publication [8]. Much of this work has been carried out by Joyce Coppock, an undergraduate student at Duke. This work will appear in the Journal {\em Chaos} (Ref.~\cite{daniels04}).

Numerical simulation results

We have carried out several numerical modeling projects, two of which were in collaboration with Prof. Lou Kondic of the New Jersey Institute of Technology. One involved generalizing the idea of granular temperature to flowing dense granular materials for large pressures and very slow speeds, a regime in which the granular temperature $T_g \propto <v^2>$ is an imperfect measure of energy fluctuations (publications [3,4,16]). We proposed an extension of T_g drawn from the idea of equipartition: $T_e = mv^2/2 + k (\delta x)^2/2$, where k is the effective spring constant associated with granular force interactions, and δx is the compression. We then compared T_e to a granular temperature drawn from energy fluctuations, and found that these two temperatures were in reasonable agreement for our model system, whereas as for the given conditions, T_g, was much too small to be representative of energy fluctuations. We are continuing work in this direction with the added collaboration of Prof.~Corey O'Hern of Yale University.

At Duke, we also carried out simulations of spherical particles rolling on a smooth frictional substrate as an experimental model for the cooling of a 2D granular gas.~\cite{kudrolli97b} Under previous support, we experimentally characterized the cooling of collections of particles on such a substrate, and found a cooling rate that considerably exceeded what would occur just from inter-particle collisions. We identified collision-induced sliding on the substrate as a key source of energy dissipation in such a system. (publications [6,13]) Our conclusion is that it is not possible to model a cooling 2D granular gas by a collection of spheres rolling on a substrate. However, this sytems does offer novel possibilities to probe granular interactions with surfaces.

Bibliography Type:	Description: (Last Updated: 03/05/2024)
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